# ENVIRONMENTAL PRODUCT DECLARATION

IN ACCORDANCE WITH EN 15804+A2 & ISO 14025

STEEL REINFORCING BAR

**EPD HUB, HUB-3476**
Published on 16.06.2025, last updated on 17.06.2025, valid until 15.06.2030

Created with One Click LCA

**Prepared by:** شركة تقنية الجودة للاعتماد Quality Techno Certification

## MANUFACTURER

| | |
|---|---|
| Manufacturer | Nuor Steel Industry Company |
| Address | Street 59, Al Kharj Industrial City 16416, Saudi Arabia |
| Contact details | +966 54 378 1868 |
| Email Address | info@nuorsteel.com |
| Website | https://nuorsteel.com/en |

## EPD STANDARDS, SCOPE AND VERIFICATION

| | |
|---|---|
| Program operator | EPD Hub, hub@epdhub.com |
| Reference standard | EN 15804+A2:2019 and ISO 14025 |
| PCR | EPD Hub Core PCR Version 1.1, 5 Dec 2023 |
| Sector | Construction product |
| Category of EPD | Third party verified EPD |
| Scope of the EPD | Cradle to gate with modules C1-C4, D |
| EPD author | Osama Alhuzebi, Quality Techno Certification |
| EPD verification | Independent verification of this EPD and data, according to ISO 14025:<br>☐ Internal verification ☑ External verification |
| EPD verifier | Sarah Curpen, as an authorized verifier acting for EPD Hub Limited. |

The manufacturer has the sole ownership, liability, and responsibility for the EPD. EPDs within the same product category but from different programs may not be comparable. EPDs of construction products may not be comparable if they do not comply with EN 15804 and if they are not compared in a building context.

## PRODUCT

| | |
|---|---|
| Product name | Steel Rebar |
| Product range | 8 mm to 32 mm |
| Place of production | Saudi Arabia |
| Period for data | January 1 to December 31, 2024 |
| Averaging in EPD | No averaging |
| Variation in GWP-fossil for A1-A3 | -% |

## ENVIRONMENTAL DATA SUMMARY

| | |
|---|---|
| **Declared unit** | 1 kg of steel rebar |
| **Declared unit mass** | 1 kg |
| **GWP-fossil, A1-A3 ($kgCO_2e$)** | 1,43+00 |
| **GWP-total, A1-A3 ($kgCO_2e$)** | 1,43E+00 |
| **Secondary material, inputs (%)** | 14,8 |
| **Secondary material, outputs (%)** | 0 |
| **Total energy use, A1-A3 (kWh)** | 4,17 |
| **Net freshwater use, A1-A3 ($m^3$)** | 0,01 |

# PRODUCT AND MANUFACTURER

## ABOUT THE MANUFACTURER

Nuor Steel Industry Company (شركة حديد نور للصناعة) is a premier manufacturer of high-quality rebar, committed to shaping the future of the steel industry through innovation, sustainability, and excellence.

Based in the Kingdom of Saudi Arabia, we are proud to align with the nation's Vision 2030 by driving industrial growth, supporting infrastructure development, and fostering environmental responsibility. Our advanced manufacturing processes and state-of-the-art technology ensure the delivery of superior steel products that meet the evolving needs of our clients and contribute to building stronger, greener foundations.

As an ISO 9001:2015, ISO 14001:2015, and ISO 45001:2015 certified company, and with full SASO licensing, Nuor Steel guarantees the highest standards of quality, environmental stewardship, and occupational health and safety. These certifications reflect our unwavering commitment to excellence and our dedication to empowering industries with durable, reliable, and sustainable steel solutions.

## PRODUCT DESCRIPTION

This EPD covers the rebar produced by Nuor Steel Industry Company. Rebar is a critical component in reinforced steel structures, essential for improving their strength and durability. Its ribbed surface design enhances its bond with concrete, ensuring a strong and reliable connection. With precise chemical composition and a carefully controlled manufacturing process.

Further information can be found at: https://nuorsteel.com/en

## PRODUCT RAW MATERIAL MAIN COMPOSITION

| Raw material category | Amount, mass % | Material origin |
|---|---|---|
| Metals | 100% | Al Kharj Industrial City - Saudi Arabia |
| Minerals | - | |
| Fossil materials | - | |
| Bio-based materials | - | |

## BIOGENIC CARBON CONTENT

Product's biogenic carbon content at the factory gate

| | |
|---|---|
| Biogenic carbon content in product, kg C | - |
| Biogenic carbon content in packaging, kg C | - |

## DECLARED UNIT

| | |
|---|---|
| Declared unit | 1 kg of steel rebar |
| Mass per declared unit | 1 kg |

## SUBSTANCES, REACH - VERY HIGH CONCERN

The product does not contain any REACH SVHC substances in amounts greater than 0,1 % (1000 ppm).

# PRODUCT LIFE-CYCLE

## SYSTEM BOUNDARY

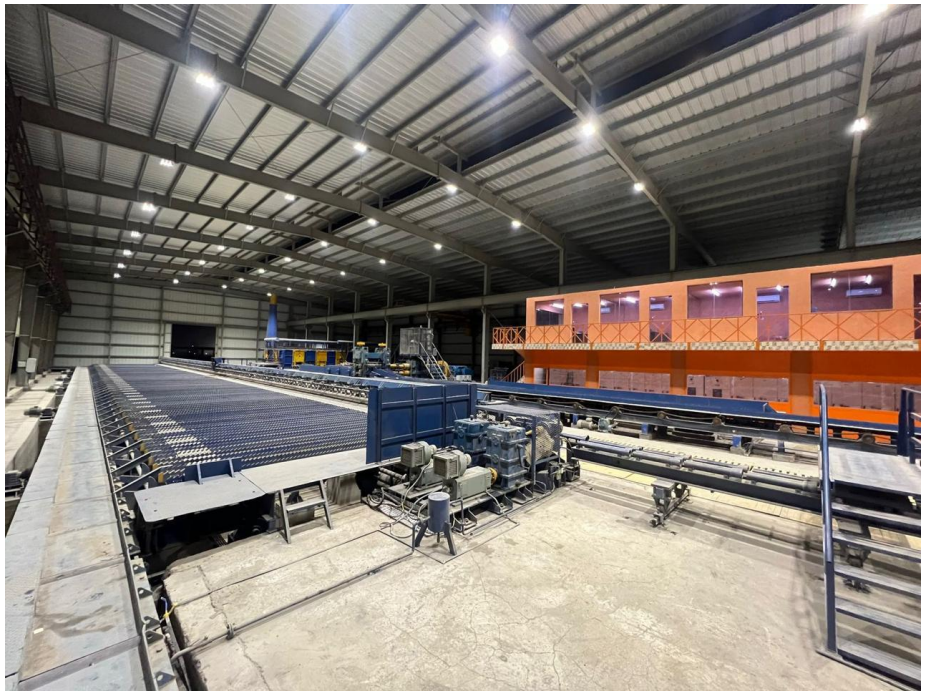
This EPD covers the life-cycle modules listed in the following table.

| Product stage | | | Assembly stage | | Use stage | | | | | | | End of life stage | | | | Beyond the system boundaries | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A1 | A2 | A3 | A4 | A5 | B1 | B2 | B3 | B4 | B5 | B6 | B7 | C1 | C2 | C3 | C4 | D | | |
| × | × | × | MND | MND | MND | MND | MND | MND | MND | MND | MND | × | × | × | × | × | | |
| Raw materials | Transport | Manufacturing | Transport | Assembly | Use | Maintenance | Repair | Replacement | Refurbishment | Operational energy use | Operational water use | Deconstruction/ demolition | Transport | Waste processing | Disposal | Reuse | Recovery | Recycling |

*Modules not declared = MND. Modules not relevant = MNR*

## MANUFACTURING AND PACKAGING (A1-A3)

The final product consists of steel rebar ranging from 8 mm to 32 mm in diameter, manufactured by Nuor Steel at its facility located in Al Kharj Industrial City. The production process begins with steel billets measuring 130×130 mm, sourced from multiple suppliers including Al Qaryan Steel Company, Power Steel Company, and Najd Steel Company. These billets have a recycled content of 38% pre-consumer and 42% post-consumer scrap. The life cycle stages from A1 to A3 cover the full cradle-to-gate process. In A1 (raw material supply), the steel billets are produced and transported to the site. In A2 (transportation), the billets are delivered to Nuor Steel's facility. A3 (manufacturing) includes reheating the billets in a furnace, followed by multi-stage rolling (roughing, intermediate, and finishing) to shape the rebar. After rolling, the bars undergo water cooling and air cooling for mechanical properties enhancement, followed by inspection, testing, bundling, and preparation for delivery. This streamlined production ensures consistency, quality, and compliance with sustainability principles.



## TRANSPORT AND INSTALLATION (A4-A5)

This EPD does not cover this phase.

## PRODUCT USE AND MAINTENANCE (B1-B7)

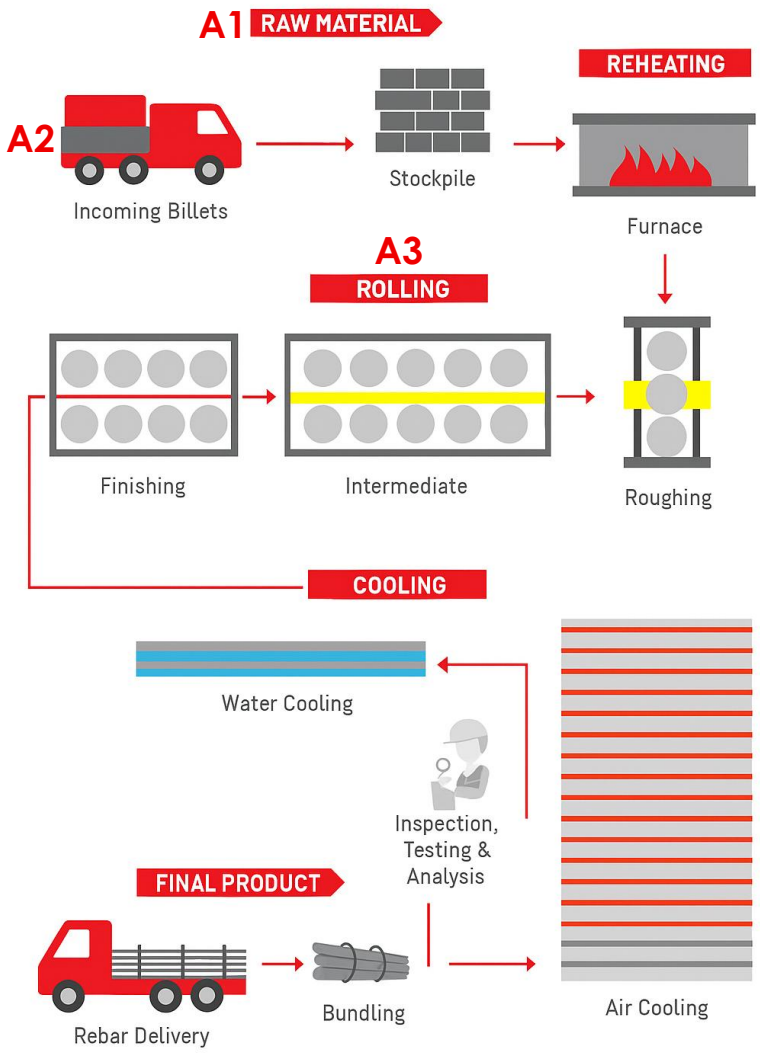
This EPD does not cover the use phase.
Air, soil, and water impacts during the use phase have not been studied.

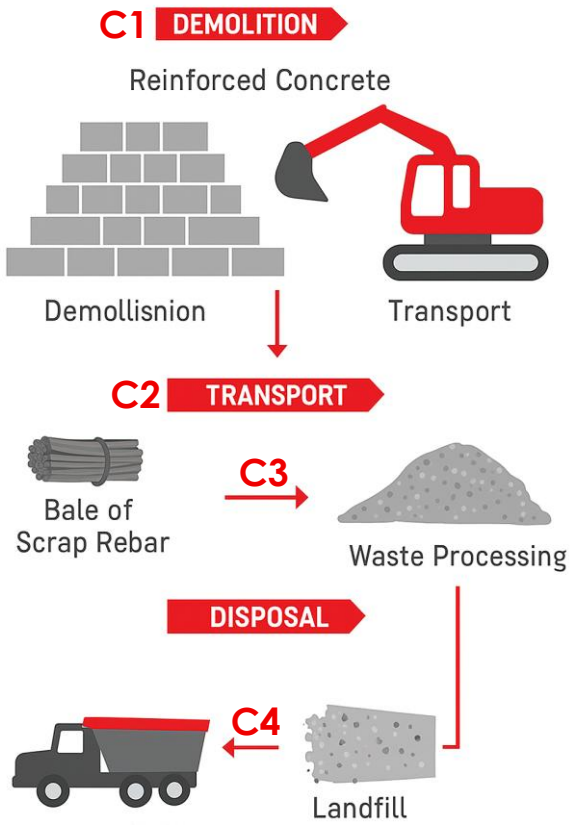
## PRODUCT END OF LIFE (C1-C4, D)

The end-of-life stage for steel reinforcement begins with the demolition of the building, during which the product is removed and initially sorted onsite. The energy consumption of building machinery used in demolition is estimated at 0.01 kWh/kg of installed product. Following this, the discarded rebar is transported to either a recycling facility or a final disposal site.

The assumed distance to the waste treatment plant is 100 km as an average. Waste processing phase includes the collection and handling of material streams designated for reuse, recycling, or energy recovery. With a service life exceeding 60 years, rebar typically reaches the end of its lifecycle alongside the building's demolition. According to the latest World Steel report 2020, 85% of steel is sent for recycling, while 15% is disposed of in landfills.

# MANUFACTURING PROCESS




**A1:** Raw material extraction

**A2:** Transportation to the factory

**A3:** Manufacturing process

**C1:** Demolition

**C2:** Transportation to the waste facility

**C3:** Waste processing

**C4:** Disposal

# LIFE-CYCLE ASSESSMENT

## CUT-OFF CRITERIA

The study does not exclude any modules or processes which are stated mandatory in the reference standard and the applied PCR. The study does not exclude any hazardous materials or substances. The study includes all major raw material and energy consumption. All inputs and outputs of the unit processes, for which data is available for, are included in the calculation. There is no neglected unit process more than 1% of total mass or energy flows. The module specific total neglected input and output flows also do not exceed 5% of energy usage or mass.

## ALLOCATION, ESTIMATES AND ASSUMPTIONS

Allocation is required if some material, energy, and waste data cannot be measured separately for the product under investigation. All allocations are done as per the reference standards and the applied PCR. In this study, allocation has been done in the following ways:

| Data type | Allocation |
|---|---|
| Raw materials | Allocated by mass |
| Packaging material | No allocation |
| Ancillary materials | Not applicable |
| Manufacturing energy and waste | Allocated by mass |

## AVERAGES AND VARIABILITY

| | |
|---|---|
| Type of average | No averaging |
| Averaging method | Not applicable |
| Variation in GWP-fossil for A1-A3 | Not applicable |

This EPD is product and factory specific and does not contain average calculations.

## LCA SOFTWARE AND BIBLIOGRAPHY

This EPD has been created using One Click LCA EPD Generator. The LCA and EPD have been prepared according to the reference standards and ISO 14040/14044. The EPD Generator uses Ecoinvent v3.10 and One Click LCA databases as sources of environmental data.

# ENVIRONMENTAL IMPACT DATA

## CORE ENVIRONMENTAL IMPACT INDICATORS – EN 15804+A2

| Impact category | Unit | A1 | A2 | A3 | A1-A3 | A4 | A5 | B1 | B2 | B3 | B4 | B5 | B6 | B7 | C1 | C2 | C3 | C4 | D |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| GWP – total[1)] | kg $CO_2$e | 1,40E+00 | 5,38E-04 | 3,05E-02 | 1,43E+00 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 3,61E-03 | 1,08E-02 | 2,31E-02 | 9,37E-04 | -1,01E+00 |
| GWP – fossil | kg $CO_2$e | 1,40E+00 | 5,38E-04 | 3,04E-02 | 1,43E+00 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 3,60E-03 | 1,08E-02 | 2,31E-02 | 9,36E-04 | -1,01E+00 |
| GWP – biogenic | kg $CO_2$e | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 |
| GWP – LULUC | kg $CO_2$e | 6,94E-04 | 2,41E-07 | 1,17E-06 | 6,96E-04 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 3,69E-07 | 4,82E-06 | 2,72E-05 | 5,35E-07 | -4,87E-04 |
| Ozone depletion pot. | kg CFC-$_{11}$e | 5,90E-09 | 7,95E-12 | 4,63E-09 | 1,05E-08 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 5,52E-11 | 1,59E-10 | 2,48E-10 | 2,71E-11 | -4,03E-09 |
| Acidification potential | mol $H^+$e | 4,86E-03 | 1,84E-06 | 1,75E-04 | 5,04E-03 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 3,25E-05 | 3,67E-05 | 2,47E-04 | 6,64E-06 | -4,04E-03 |
| EP-freshwater[2)] | kg Pe | 4,77E-04 | 4,19E-08 | 3,92E-07 | 4,77E-04 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 1,04E-07 | 8,38E-07 | 1,25E-05 | 7,70E-08 | -3,72E-04 |
| EP-marine | kg Ne | 1,12E-03 | 6,03E-07 | 3,39E-05 | 1,15E-03 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 1,51E-05 | 1,21E-05 | 5,49E-05 | 2,53E-06 | -8,64E-04 |
| EP-terrestrial | mol Ne | 1,19E-02 | 6,56E-06 | 3,56E-04 | 1,23E-02 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 1,65E-04 | 1,31E-04 | 6,19E-04 | 2,76E-05 | -9,34E-03 |
| POCP ("smog")[3)] | kg NMVOCe | 4,45E-03 | 2,70E-06 | 1,02E-04 | 4,55E-03 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 4,93E-05 | 5,41E-05 | 1,82E-04 | 9,90E-06 | -3,08E-03 |
| ADP-minerals & metals[4)] | kg Sbe | 1,08E-06 | 1,50E-09 | 1,29E-08 | 1,09E-06 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 1,29E-09 | 3,00E-08 | 1,36E-06 | 1,49E-09 | -7,65E-06 |
| ADP-fossil resources | MJ | 1,40E+01 | 7,81E-03 | 4,30E-01 | 1,44E+01 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 4,72E-02 | 1,56E-01 | 2,73E-01 | 2,30E-02 | -9,75E+00 |
| Water use[5)] | $m^3$e depr. | 6,50E-01 | 3,86E-05 | 1,62E-03 | 6,51E-01 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 1,18E-04 | 7,72E-04 | 4,31E-03 | 6,63E-05 | -2,62E-01 |

*1) GWP = Global Warming Potential; 2) EP = Eutrophication potential. Required characterisation method and data are in kg P-eq. Multiply by 3,07 to get PO4e; 3) POCP = Photochemical ozone formation; 4) ADP = Abiotic depletion potential; 5) EN 15804+A2 disclaimer for Abiotic depletion and Water use and optional indicators except Particulate matter and Ionizing radiation, human health. The results of these environmental impact indicators shall be used with care as the uncertainties on these results are high or as there is limited experience with the indicator.*

## ADDITIONAL (OPTIONAL) ENVIRONMENTAL IMPACT INDICATORS – EN 15804+A2

| Impact category | Unit | A1 | A2 | A3 | A1-A3 | A4 | A5 | B1 | B2 | B3 | B4 | B5 | B6 | B7 | C1 | C2 | C3 | C4 | D |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Particulate matter | Incidence | 1,18E-07 | 5,39E-11 | 1,13E-09 | 1,19E-07 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 9,25E-10 | 1,08E-09 | 3,45E-09 | 1,51E-10 | -9,04E-08 |
| Ionizing radiation[6)] | kBq U235e | 2,80E-02 | 6,80E-06 | 8,20E-04 | 2,88E-02 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 2,09E-05 | 1,36E-04 | 9,79E-04 | 1,44E-05 | -2,06E-02 |
| Ecotoxicity (freshwater) | CTUe | 3,93E+00 | 1,10E-03 | 1,75E-01 | 4,11E+00 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 2,60E-03 | 2,21E-02 | 1,58E-01 | 1,93E-03 | -3,13E+00 |
| Human toxicity, cancer | CTUh | 1,48E-09 | 8,88E-14 | 6,30E-12 | 1,49E-09 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 3,71E-13 | 1,78E-12 | 1,85E-11 | 1,73E-13 | -1,14E-09 |
| Human tox. non-cancer | CTUh | 5,71E-09 | 5,06E-12 | 8,79E-11 | 5,80E-09 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 5,87E-12 | 1,01E-10 | 1,18E-09 | 3,97E-12 | -8,75E-09 |
| SQP[7)] | - | 3,13E+00 | 7,87E-03 | 4,34E-03 | 3,15E+00 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 3,30E-03 | 1,57E-01 | 5,16E-01 | 4,52E-02 | -3,20E+00 |

6) EN 15804+A2 disclaimer for Ionizing radiation, human health. This impact category deals mainly with the eventual impact of low-dose ionizing radiation on human health of the nuclear fuel cycle. It does not consider effects due to possible nuclear accidents, occupational exposure nor due to radioactive waste disposal in underground facilities. Potential ionizing radiation from the soil, from radon and from some construction materials is also not measured by this indicator; 7) SQP = Land use related impacts/soil quality.

## USE OF NATURAL RESOURCES

| Impact category | Unit | A1 | A2 | A3 | A1-A3 | A4 | A5 | B1 | B2 | B3 | B4 | B5 | B6 | B7 | C1 | C2 | C3 | C4 | D |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Renew. PER as energy[8)] | MJ | 5,59E-01 | 1,07E-04 | 1,47E-03 | 5,61E-01 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 2,99E-04 | 2,14E-03 | 4,24E-02 | 2,22E-04 | -9,61E-01 |
| Renew. PER as material | MJ | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 |
| Total use of renew. PER | MJ | 5,59E-01 | 1,07E-04 | 1,47E-03 | 5,61E-01 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 2,99E-04 | 2,14E-03 | 4,24E-02 | 2,22E-04 | -9,61E-01 |
| Non-re. PER as energy | MJ | 1,40E+01 | 7,81E-03 | 4,30E-01 | 1,44E+01 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 4,72E-02 | 1,56E-01 | 2,73E-01 | 2,30E-02 | -9,75E+00 |
| Non-re. PER as material | MJ | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 |
| Total use of non-re. PER | MJ | 1,40E+01 | 7,81E-03 | 4,30E-01 | 1,44E+01 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 4,72E-02 | 1,56E-01 | 2,73E-01 | 2,30E-02 | -9,75E+00 |
| Secondary materials | kg | 1,48E-01 | 3,32E-06 | 5,41E-06 | 1,48E-01 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 1,96E-05 | 6,65E-05 | 3,16E-04 | 5,78E-06 | -8,72E-02 |
| Renew. secondary fuels | MJ | 5,83E-05 | 4,22E-08 | 4,48E-09 | 5,84E-05 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 5,12E-08 | 8,45E-07 | 1,43E-05 | 1,20E-07 | -1,16E-04 |
| Non-ren. secondary fuels | MJ | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 |
| Use of net fresh water | $m^3$ | 1,44E-02 | 1,15E-06 | 1,98E-04 | 1,46E-02 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 3,12E-06 | 2,31E-05 | 1,19E-04 | 2,39E-05 | -5,67E-03 |

*8) PER = Primary energy resources.*

## END OF LIFE – WASTE

| Impact category | Unit | A1 | A2 | A3 | A1-A3 | A4 | A5 | B1 | B2 | B3 | B4 | B5 | B6 | B7 | C1 | C2 | C3 | C4 | D |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Hazardous waste | kg | 2,66E-01 | 1,32E-05 | 1,03E-04 | 2,66E-01 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 5,25E-05 | 2,65E-04 | 2,13E-03 | 2,54E-05 | -4,09E-01 |
| Non-hazardous waste | kg | 2,68E+00 | 2,45E-04 | 4,44E-03 | 2,68E+00 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 7,15E-04 | 4,90E-03 | 5,98E-02 | 5,80E-04 | -2,10E+00 |
| Radioactive waste | kg | 6,80E-06 | 1,67E-09 | 1,31E-06 | 8,12E-06 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 5,12E-09 | 3,33E-08 | 2,41E-07 | 3,52E-09 | -5,11E-06 |

## END OF LIFE – OUTPUT FLOWS

| Impact category | Unit | A1 | A2 | A3 | A1-A3 | A4 | A5 | B1 | B2 | B3 | B4 | B5 | B6 | B7 | C1 | C2 | C3 | C4 | D |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Components for re-use | kg | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 |
| Materials for recycling | kg | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 |
| Materials for energy rec | kg | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 |
| Exported energy | MJ | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 |
| Exported energy – Electricity | MJ | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 |
| Exported energy – Heat | MJ | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 |

## ENVIRONMENTAL IMPACTS – EN 15804+A1, CML

| Impact category | Unit | A1 | A2 | A3 | A1-A3 | A4 | A5 | B1 | B2 | B3 | B4 | B5 | B6 | B7 | C1 | C2 | C3 | C4 | D |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Global Warming Pot. | kg $CO_2e$ | 1,39E+00 | 5,35E-04 | 2,98E-02 | 1,42E+00 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 3,59E-03 | 1,07E-02 | 2,30E-02 | 9,28E-04 | -1,00E+00 |
| Ozone depletion Pot. | kg $CFC_{-11}e$ | 5,24E-09 | 6,34E-12 | 3,63E-09 | 8,88E-09 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 4,37E-11 | 1,27E-10 | 2,06E-10 | 2,15E-11 | -3,59E-09 |
| Acidification | kg $SO_2e$ | 3,92E-03 | 1,40E-06 | 1,43E-04 | 4,06E-03 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 2,29E-05 | 2,80E-05 | 1,99E-04 | 4,91E-06 | -3,28E-03 |
| Eutrophication | kg $PO_4^{3}e$ | 6,88E-04 | 3,41E-07 | 1,32E-05 | 7,02E-04 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 5,34E-06 | 6,83E-06 | 2,83E-05 | 1,56E-06 | -5,48E-04 |
| POCP (“smog”) | kg $C_2H_4e$ | 5,87E-04 | 1,25E-07 | 5,73E-06 | 5,93E-04 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 1,71E-06 | 2,50E-06 | 1,18E-05 | 4,65E-07 | -4,36E-04 |
| ADP-elements | kg Sbe | 9,86E-07 | 1,46E-09 | 1,28E-08 | 1,00E-06 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 1,26E-09 | 2,93E-08 | 1,36E-06 | 1,46E-09 | -7,59E-06 |
| ADP-fossil | MJ | 1,36E+01 | 7,70E-03 | 4,30E-01 | 1,40E+01 | MND | MND | MND | MND | MND | MND | MND | MND | MND | 4,68E-02 | 1,54E-01 | 2,57E-01 | 2,28E-02 | -9,43E+00 |

# VERIFICATION STATEMENT

## VERIFICATION PROCESS FOR THIS EPD

This EPD has been verified in accordance with ISO 14025 by an independent, third-party verifier by reviewing results, documents and compliancy with reference standard, ISO 14025 and ISO 14040/14044, following the process and checklists of the program operator for:

- This Environmental Product Declaration
- The Life-Cycle Assessment used in this EPD
- The digital background data for this EPD

Why does verification transparency matter? Read more online
This EPD has been generated by One Click LCA EPD generator, which has been verified and approved by the EPD Hub.

## THIRD-PARTY VERIFICATION STATEMENT

I hereby confirm that, following detailed examination, I have not established any relevant deviations by the studied Environmental Product Declaration (EPD), its LCA and project report, in terms of the data collected and used in the LCA calculations, the way the LCA-based calculations have been carried out, the presentation of environmental data in the EPD, and other additional environmental information, as present with respect to the procedural and methodological requirements in ISO 14025:2010 and reference standard.

I confirm that the company-specific data has been examined as regards plausibility and consistency; the declaration owner is responsible for its factual integrity and legal compliance.

I confirm that I have sufficient knowledge and experience of construction products, this specific product category, the construction industry, relevant standards, and the geographical area of the EPD to carry out this verification.

I confirm my independence in my role as verifier; I have not been involved in the execution of the LCA or in the development of the declaration and have no conflicts of interest regarding this verification.

Sarah Curpen, as an authorized verifier acting for EPD Hub Limited.
16.06.2025